

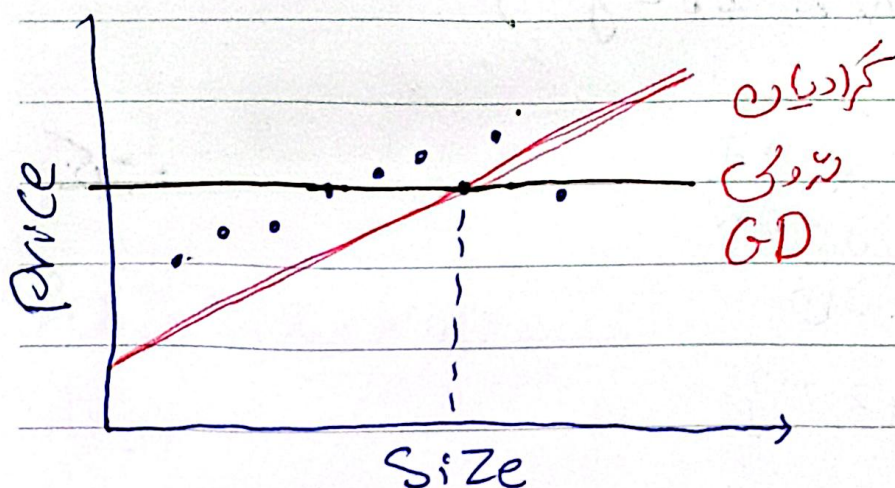
1 «دروس مصنوعی» در استار فوری»

2 «در جزوه طبعی» پایانی

$$f(x) = wx + b$$

$$f: x \rightarrow y$$

Linear Regression



$$D_m = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$$

$$f(x^{(1)}) - y^{(1)}$$

$$f(x^{(i)}) - y^{(i)}$$

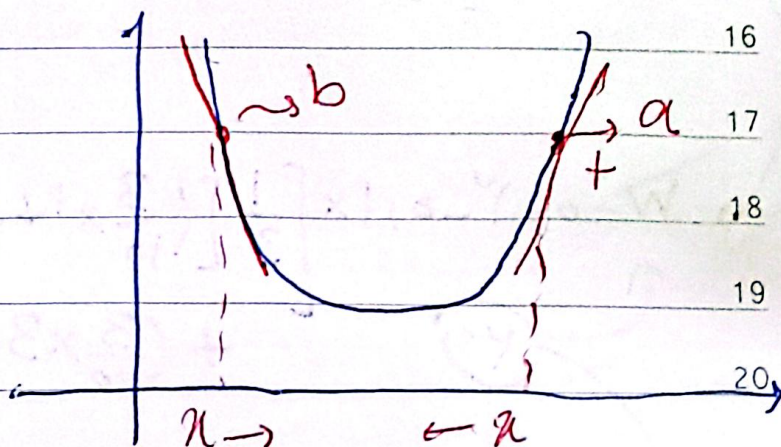
$$f(x^{(m)}) - y^{(m)}$$

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})^2$$

MSE

$$x - a$$

$$x - b$$



$$w \leftarrow w - \frac{\partial J}{\partial w}$$

MRNOTE

$$b \leftarrow b - \frac{\partial J}{\partial b}$$

$$\frac{\partial J}{\partial w} = \frac{1}{2M} \sum_{i=1}^m 2(x^{(i)})(wx^{(i)} + b - y^{(i)})$$

$$= \frac{1}{m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) x^{(i)}$$

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})$$

$$w \leftarrow w - \alpha \frac{\partial J}{\partial w} \quad \text{نکته}$$

$$b \leftarrow b - \alpha \frac{\partial J}{\partial b}$$

$$\{(1,1), (2,1), (3,2)\}$$

(داده)

$$\alpha = 0,1$$

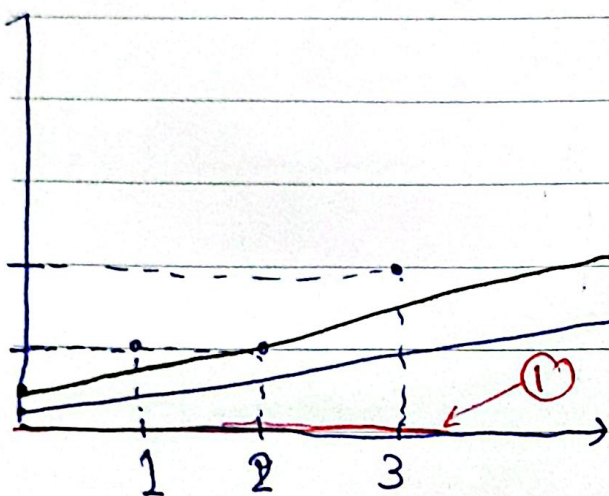
$$w = 0, b = 0$$

$$w = 0 - 0,1 \times \left[\frac{1}{3} [(0 \times 1 + 0 - 1) \times 1 + (0 \times 2 + 0 - 1) \times 2 + (0 \times 3 + 0 - 2) \times 3] \right] = 0,3$$

$$b = 0 - 0,1 \times \left[\frac{1}{3} [(0 \times 1 + 0 - 1) + (0 \times 2 + 0 - 1) + (0 \times 3 + 0 - 2)] \right] = \frac{4}{30} = \frac{2}{15}$$

$$w = 0,3 - 0,1 \times \left[\frac{1}{3} \left[\left(\frac{3}{10} \times 1 + \frac{2}{15} - 1 \right) \times 1 + \left(\frac{3}{10} \times 2 + \frac{2}{15} - 1 \right) \times 2 + \left(\frac{3}{10} \times 3 + \frac{2}{15} - 2 \right) \times 3 \right] \right] = 0,44$$

$$b = 0,13 - 0,1 \times \left[\frac{1}{3} \left[\left(\frac{3}{10} \times 1 + \frac{2}{15} - 1 \right) + \left(\frac{3}{10} \times 2 + \frac{2}{15} - 1 \right) + \left(\frac{3}{10} \times 3 + \frac{2}{15} - 2 \right) \right] \right] = 0,19$$



در صورتی که چند متغیر داشته باشیم:

$$f(x) = w_1 x_1 + w_2 x_2 + \dots + b$$

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \begin{matrix} \rightarrow x^{(i)} \\ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \end{matrix}$$

در صورتی که متغیرها غیر فکلی (توان دار باشند):

یک متغیر دیگر (ایجاد می کنیم که این متغیر جدید توانی از متغیر اول

است: $x_2 = x_1^2$ و حال با این تبدیل متغیرها باز

فکلی می شوند و ما شد روش قبل حل می کنیم.

$$w_1 x_1 \rightarrow w_2 x_2 \quad x_2 = x_1^2$$

$$3 \rightarrow 9, 5 \rightarrow 25, 2 \rightarrow 4$$

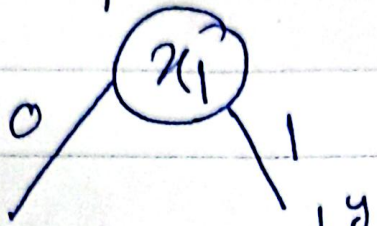
x_1	x_2	x_3	y
1	1	0	1
1	0	1	1
0	0	1	0
0	1	1	1
1	0	0	0
1	1	0	1
0	0	0	0

مثالی از درخت تصمیم:

$$P_1 = \frac{4}{7}$$

$$P_0 = \frac{3}{7}$$

$$H\left(\frac{4}{7}\right) = -\frac{4}{7} \log_2 \frac{4}{7} - \frac{3}{7} \log_2 \frac{3}{7} = 0.98$$



				y
0	0	1	0	0
0	1	1	1	1
0	0	0	0	0

$$P_1 = \frac{1}{3}$$

$$P_0 = \frac{2}{3}$$

				y
1	1	0	1	1
1	0	1	1	1
1	0	0	0	0
1	1	1	1	1

$$P_1 = \frac{3}{4}$$

$$P_0 = \frac{1}{4}$$

$$H\left(\frac{1}{3}\right) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.91$$

$$H\left(\frac{3}{4}\right) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.81$$

$$IG(x_1) = H\left(\frac{4}{7}\right) - \left(\frac{3}{7} \times H\left(\frac{1}{3}\right) + \frac{4}{7} H\left(\frac{3}{4}\right)\right)$$

$$= 0.98 - \left(\frac{3}{7} \times 0.91 + \frac{4}{7} \times 0.81\right) = 0.12$$

$$H\left(\frac{4}{7}\right) = 0.98$$



				y
1	0	1	1	1
0	0	1	0	0
1	0	0	0	0
0	0	0	0	0

$$P_1 = \frac{1}{4}$$

$$P_0 = \frac{3}{4}$$

$$P_1 = 1$$

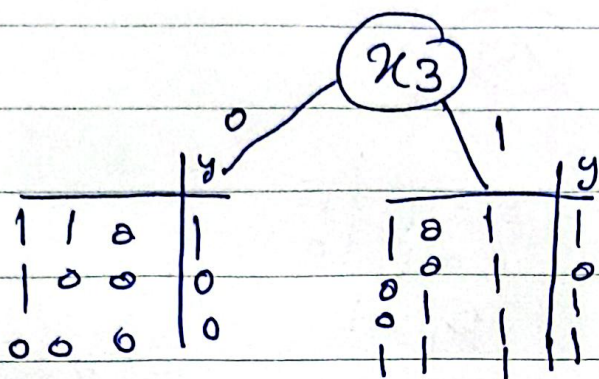
$$P_0 = 0$$

$$H(1) = -1 \log_2 1 - 0 \log_2 0 = 0$$

$$H\left(\frac{1}{4}\right) = -\frac{1}{4} \log_2 \frac{1}{4} - \frac{3}{4} \log_2 \frac{3}{4} = 0.81$$

$$IG(x_2) = 0.98 - \left(\frac{4}{7} \times 0.81 + \frac{3}{7} \times 0\right) = 0.51$$

$$H\left(\frac{4}{7}\right) = 0,98$$



$$P_1 = \frac{1}{3}$$

$$P_0 = \frac{2}{3}$$

$$P_1 = \frac{3}{4}$$

$$P_0 = \frac{1}{4}$$

$$H\left(\frac{1}{3}\right) = 0,91$$

$$H\left(\frac{3}{4}\right) = 0,81$$

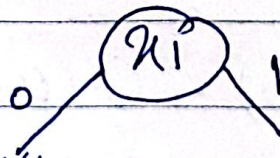
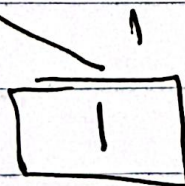
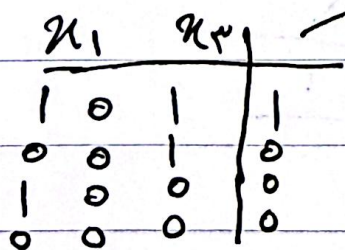
$$IG(x_3) = 0,98 - \left(\frac{3}{7} \times 0,91 + \frac{4}{7} \times 0,81\right) = 0,12$$

$$IG(x_1) = 0,12$$

$$IG(x_3) = 0,12$$

$$IG(x_2) = 0,51$$

$$H\left(\frac{1}{4}\right) = 0,81$$



$$IG(x_1) = 0,81$$

$$-\left(\frac{2}{4} \times 0 + \frac{2}{4} \times 1\right) = 0,31$$

$$P_1 = 0$$

$$P_0 = 1$$

$$H(0) = 0$$

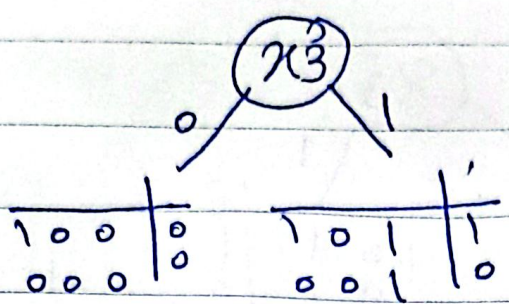
"5"

$$P_1 = \frac{1}{2}$$

$$P_0 = \frac{1}{2}$$

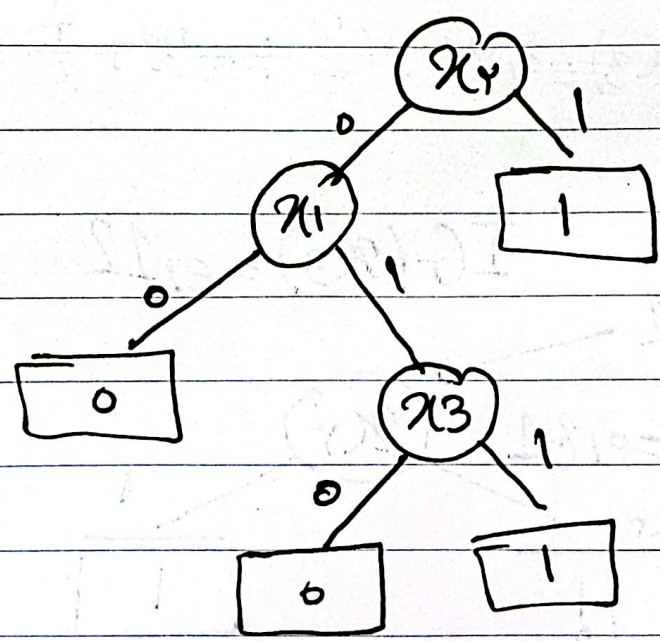
$$H\left(\frac{1}{2}\right) = 1$$

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22



$IG(x_3) = 0.31$
 $P_1 = 0$
 $P_0 = 1$
 $H(0) = 0$
 $P_1 = \frac{1}{2}$
 $P_0 = \frac{1}{2}$
 $H(\frac{1}{2}) = 1$

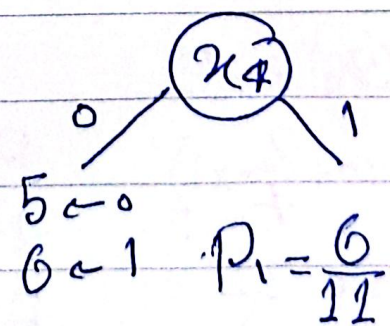
هر درخت در نهایت:



$Gini(x_i) = G(\text{Parent}) - \left[\frac{n_L}{n} G(L) + \frac{n_R}{n} G(R) \right]$

$G(\text{node}) = 1 - \sum p_i^2$

$1 - \left[\left(\frac{5}{11} \right)^2 + \left(\frac{6}{11} \right)^2 \right]$



و بن

$P_1 = \frac{6}{11}$
 $P_0 = \frac{5}{11}$

$$\text{Classification Error} = E(\text{Parent}) - \left[\frac{n_L}{n} E(L) + \frac{n_R}{n} E(R) \right]$$

$$E(\text{node}) = 1 - \max P_i = 1 - \frac{6}{11} = \frac{5}{11}$$

$$IG(X) = H(\text{Parent}) - \left[\frac{n_L}{n} H(L) + \frac{n_R}{n} H(R) \right]$$

$$f(x) = w_2 x_2 + w_1 x_1 + b$$

$$w_1 = 0$$

$$w_2 = 0 \quad \alpha = 0.1$$

$$b = 0$$

$x_1^{(1)}$	$x_1^{(2)}$	y
-1	1	1
0	0	0
1	1	1
2	2	2

$$b = 0 - \frac{0.1}{\epsilon} \times [(0-1) + (0-0) + (0-1) + (0-2)] = 0.15$$

$$w_1 = 0 - \frac{0.1}{\epsilon} \times [(0-1) \times -1 + (0-0) \times 0 + (0-1) \times 1 + (0-2) \times 2] = 0.1$$

$$w_2 = 0 - \frac{0.1}{\epsilon} \times [(0-1) \times 1 + (0-0) \times 0 + (0-1) \times 1 + (0-2) \times 2] = 0.15$$

$$b = 0.15 - \frac{0.1}{\epsilon} \times [(0.15-1) + (0.15-0) + (0.15-1) + (0.15-2)] = 0.12$$

$$w_1 = 0.1 - \frac{0.1}{\epsilon} \times [(0.15-1) \times -1 + (0.15-0) \times 0 + (0.15-1) \times 1 + (0.15-2) \times 2] = 0.12$$

$$w_2 = 0.15 - \frac{0.1}{\epsilon} \times [(0.15-1) \times 1 + (0.15-0) \times 0 + (0.15-1) \times 1 + (0.15-2) \times 2] = 0.12$$

$$f(x) = 0.635 x_1 + 0.27 x_2 + 0.12$$

$$f(x) = 0,62x_0 + 0,12x_1 + 0,15$$

2

$$f(1) = 1,15$$

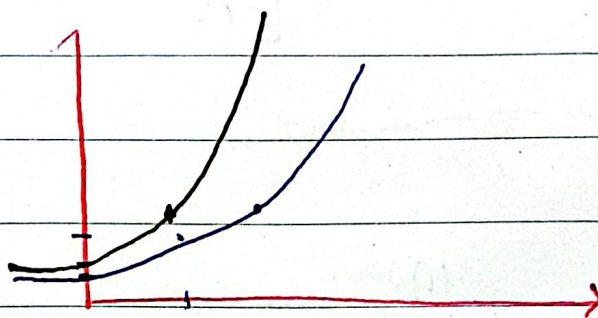
$$f(1) = 0,11$$

$$f(2) = 2,29$$

$$f(2) = 1,29$$

$$f(0) = 0,21$$

$$f(0) = 0,15$$



10

$$b = b - \alpha \frac{\partial J}{\partial b} \quad \frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)})$$

12

$$w_j = w_j - \alpha \frac{\partial J}{\partial w_j} \quad \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (f(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

14

15

16

17

18

19

20

21

22

16m